ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY

PAPER 3

6031/3

NOVEMBER 2018 SESSION

2 hours 30 minutes

Additional materials:
Answer paper
Data Booklet
Mathematical tables and/or electronic calculator

TIME: 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from Section C and **one** question from Section D.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 11 printed pages and 1 blank page.

Section A

Answer any two questions from this section.

1 (a) **Fig.1.1** shows the mass spectrum of a sample of an element Y.

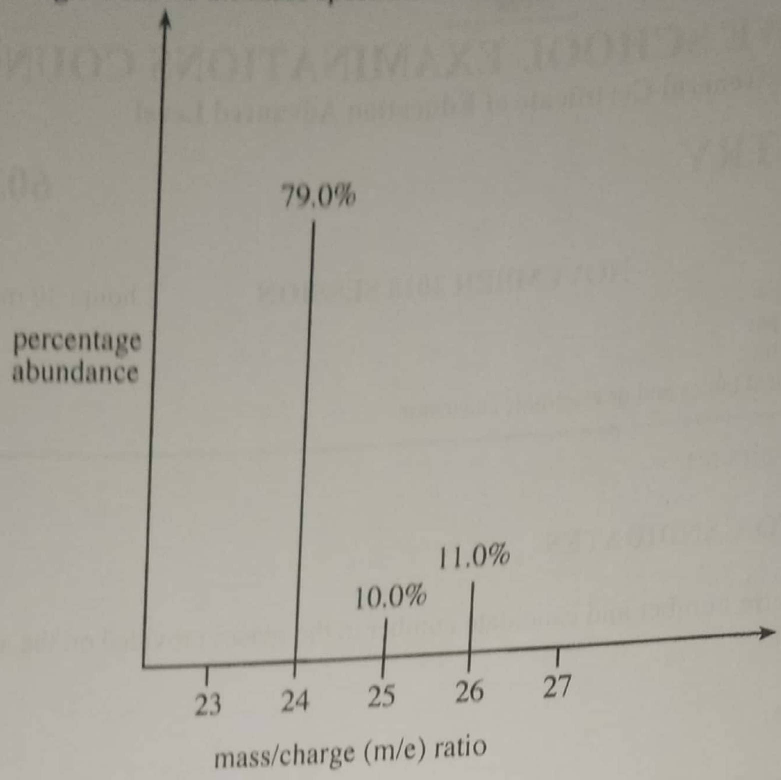


Fig.1.1

(i) Define the term *relative atomic mass.*

(ii) Calculate the relative atomic mass of Y.

(iii) Deduce the full electronic configuration of Y.

(iv) Identify the species responsible for the tallest peak.

(v) Explain why there are three different peaks in the mass spectrum of Y. [6]

(b) (i) Define the term *ionic product of water*, K_w.

(ii) Explain, using K_w, why at 25°C water has a pH of 7. [4]

(c) An aqueous buffer solution containing 0.055 moles of weak acid, HA, and 0.025 moles of NaA in 100 cm^3 of solution has a pH of 4.20.
[K_a of the acid, HA, is 2.87×10^{-5} $moldm^{-3}$ at 25°C]

Calculate the pH of the solution formed when 10 cm^3 of 0.130 $moldm^{-3}$ sodium hydroxide are added to 100 cm^3 of the buffer solution. [5]

[Total:15]

2 (a) **Table 2.1** shows enthalpy changes of some chemical processes.

Table 2.1

standard enthalpy of atomisation of magnesium	$+150\ kJmol^{-1}$
first electron affinity of oxygen	$-142\ kJmol^{-1}$
standard enthalpy of formation of magnesium oxide	$-602\ kJmol^{-1}$
lattice enthalpy of magnesium oxide	$-3\ 888\ kJmol^{-1}$

(i) Draw a Born-Haber cycle for the formation of magnesium oxide.

(ii) Calculate the second electron affinity of oxygen.

(iii) Explain why

1. the second electron affinity of oxygen is endothermic.

2. magnesium oxide is used as a refractory lining in furnaces.

[8]

(b) In an experiment to determine the rate of reaction between $S_2O_8^{2-}$ and I^- ions, in aqueous solution, the data in **Table 2.2** was obtained.

Table 2.2

experiment number	$[S_2O_8^{2-}]$/ **moldm^{-3}**	**[I$^-$] moldm^{-3}**	**initial rate/ moldm^{-3}s^{-1}**
1	0.080	0.034	2.2×10^{-4}
2	0.080	0.017	1.1×10^{-4}
3	0.160	0.017	2.2×10^{-4}

(i) Write a balanced ionic equation for the reaction.

(ii) Determine the rate equation.

(iii) Calculate the rate constant, *k*.

(iv) Suggest an experimental method than can be used to determine the rate of this reaction. [7]

[Total:15]

3 (a) (i) Write the overall equation of the reaction in a hydrogen-oxygen fuel cell operating under alkaline conditions.

(ii) Calculate the e.m.f. of the cell.

(iii) State **one** advantage of fuel cells over lead-acid accumulators.

(iv) Explain why the e.m.f. of an acidic hydrogen-oxygen fuel cell is exactly the same as that of an alkaline fuel cell.

(v) State **one** advantage, other than cost, of using porous coated electrodes in fuel cells. [5]

(b) A hydrogen-oxygen fuel cell is required to produce a current of 0.03 A for 50 days.

Calculate the mass of hydrogen gas that will be needed. [3]

(c) Sulphur dioxide present in an air sample was dissolved in water. The solution was then titrated against acidified potassium manganate (VII) solution. 7.40 cm^3 of 3.16 gdm^{-3}potassium manganate (VII) solution was required to reach the endpoint.

(i) Construct a balanced chemical equation for the reaction.

(ii) Determine the mass of sulphur dioxide in the air sample. [7]

[Total:15]

Section B

*Answer **one** question from this section.*

4 (a) Describe and explain

1. how the melting points of Group (II) elements vary down the group,

2. the trend in the decomposition temperatures of Group (II) nitrates.

[6]

(b) Hydrogen sulphide is produced when concentrated sulphuric acid is added to solid sodium iodide but sulphur dioxide is produced when concentrated sulphuric acid is added to solid sodium bromide.

(i) State, with reasons, the halide ion that has a greater reducing power.

(ii) Write an equation for the formation of

1. sulphur dioxide,

2. hydrogen sulphide.

(iii) In addition to sulphur dioxide and hydrogen sulphide, another reduction product is formed in the reactions in **(b)**.

1. Name the reduction product.

2. Deduce a half equation for its formation.

[9]

[Total:15]

5 (a) (i) Compare the solubilities of MgO and CaO in water.

(ii) Explain why Group (II) metals are suitable for use in fireworks.

[4]

(b) Explain the

(i) variation in volatility of Group (VII) elements,

(ii) trend in the thermal stabilities of Group (VII) hydrides.

[6]

(c) LoSalt is a mixture of sodium chloride and potassium chloride.

Describe and explain how the total concentration of chloride in LoSalt can be determined.

[5]

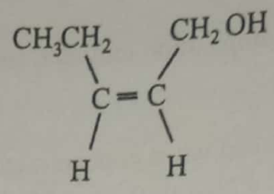
[Total:15]

[Turn over

Section C

Answer any two questions from this section.

6 **Fig.6.1** shows the structure of an organic compound **C**.



compound **C**

Fig.6.1

(a) Draw the structure of the organic product formed when **C** reacts with

(i) a solution of bromine in tetrachloromethane,

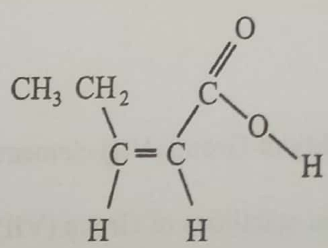
(ii) CH_3COOH in the presence of an acid catalyst. [2]

(b) Compounds **D** and **E** were formed when **C** was reacted with HBr.

(i) Draw structures of **D** and **E**.

(ii) Explain how **D** and **E** are related. [3]

(c) **Fig.6.2** shows the structure of a compound **F** that can be prepared from compound **C**.



F

Fig.6.2

(i) Suggest reagents for the conversion of **C** to **F**.

(ii) Identify the intermediate for the reaction in **c (i)**.

(iii) Describe how the intermediate can be removed from the reaction mixture.

(iv) State the type of reaction that produces the intermediate. [4]

(d) (i) Describe the bonding in benzene in terms of sigma and pi bonds.

(ii) Outline the mechanism for the reaction of benzene with $Cl_{2(g)}$ in the presence of $FeCl_3$ catalyst.

[6]

[Total:15]

7 **Fig.7.1** shows a reaction pathway of a polymer.

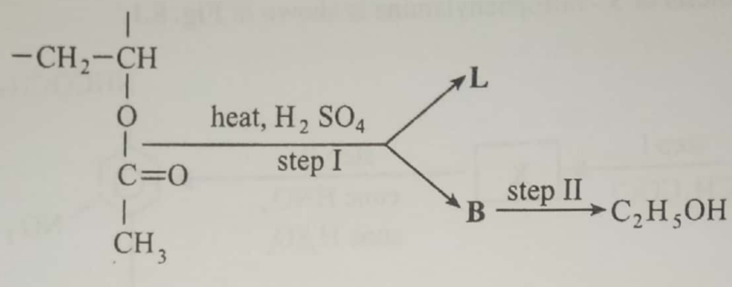


Fig.7.1

(a) (i) Draw the structural formula of

1. the monomer from which the polymer was made,
2. **L**,
3. **B**.

(ii) Identify any **two** functional groups in the monomer.

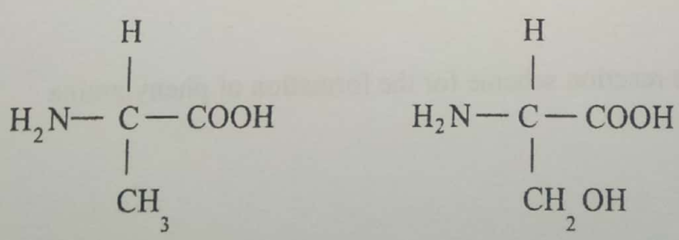
(iii) The polymer was reacted with excess sodium hydroxide to form another polymer.

Draw the repeat unit of the new polymer.

(iv) Give reagents and conditions for step II.

[9]

(b) **Fig.7.2** shows the structure of two organic comppounds, **W** and **Z**,



W **Z**

Fig.7.2

(i) Draw the structure of

1. the compound formed when **W** and **Z** react in aqueous solution.
2. W in a neutral aqueous solution.

(ii) Explain the amphoteric behaviour of **W**.

(iii) State, with reasons, which compound, **W** or **Z**, will form enantiomers.

[6]

[Total:15]

8 (a) The synthesis of 3 - nitrophenylamine is shown in **Fig. 8.1**.

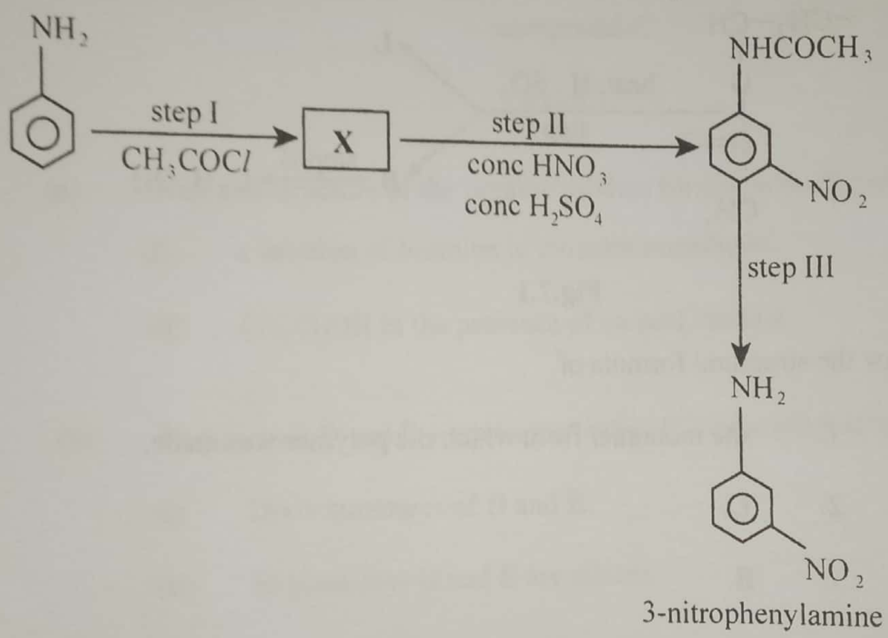


Fig. 8.1

(i) Draw the structure of **X**.

(ii) Name and outline a mechanism for step II.

(iii) Deduce the type of reaction for step III.

(iv) State the reagents and conditions for step III.

[7]

(b) Describe a two step reaction scheme for the formation of phenylamine from benzene. [3]

(c) An organic compound of molecular formula C_4H_8O gave an orange precipitate in the presence of 2,4- DPNH. The compound did not give an observable change when treated with acidified $K_2Cr_2O_7$.

Deduce, with reasons, the

(i) functional group(s) present in the compound,

(ii) possible structural formula of the compound.

[5]

[Total:15]

Section D

Answer **one** question from this section.

9 (a) Outline the **six** steps involved in the production of a genetic fingerprint. [3]

(b) One of the steps in (a) involves the use of a radioactive isotope.

(i) Name the isotope used.

(ii) State **one** reason why this isotope is chosen. [2]

(c) State any **two** factors that affect the speed at which amino acids move towards an electrode during electrophoresis. [2]

(d) **Fig.9.1** shows the results of the electrophoresis of a mixture of four acids, K, L, M and N, at a known pH.

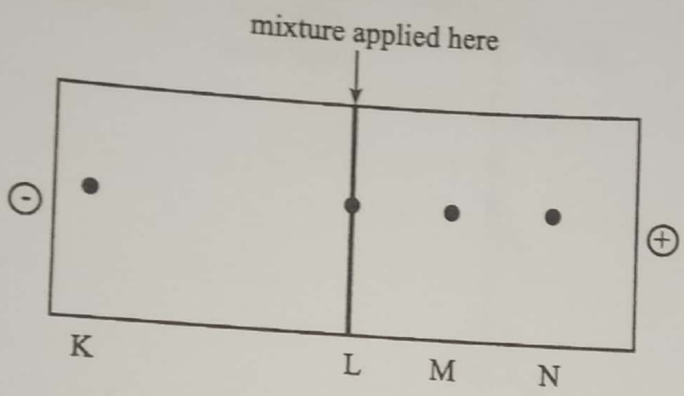


Fig.9.1

The amino acids that were present in the mixture are shown in **Fig.9.2**.

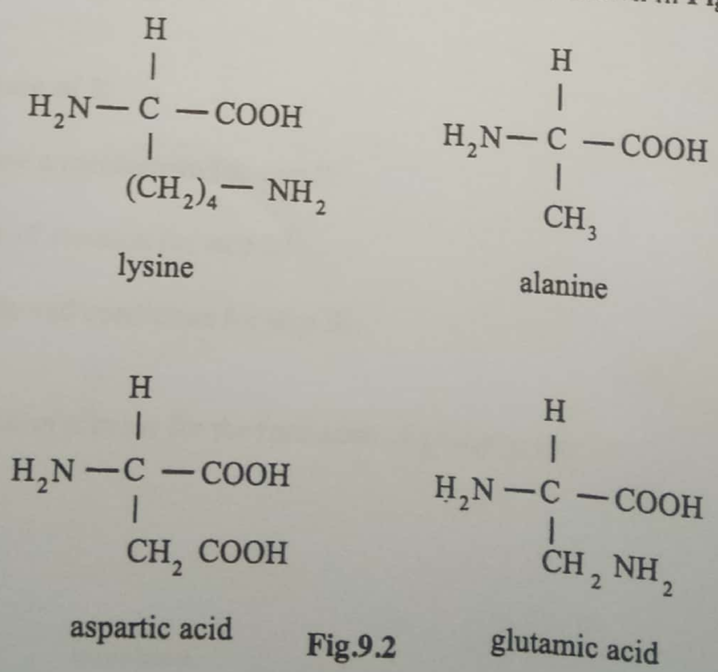


Fig.9.2

(i) Identify, with reasons, the amino acids K, L, M and N. [4]

(ii) In each case draw the structure of the amino acid as it appears at this pH. [4]

[Total:15]

10 (a) Modern catalytic converters consist of compacted nano particles coated with Pt/Rh catalyst.

(i) Explain the advantage of making the catalytic converter using nano particles.

(ii) State the type of catalysis that occurs in the catalytic converter.

(iii) Explain how Pt/ Rh function as a catalyst.

(iv) Write an equation for the reaction that occurs in the catalytic converter. [8]

(b) (i) State the origin of sulphur in fossil fuels.

(ii) Write an equation for the formation of sulphur dioxide in thermal power stations.

(iii) Describe the detrimental effects of sulphur dioxide in the atmosphere.

(iv) Explain with the aid of equations, two methods used to minimise the release of sulphur dioxide into the atmosphere at a thermal power station. [7]

[Total:15]